

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

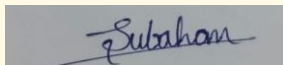
CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO1	Assessment:	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	Total Marks:	25
CO1 Statement:	Apply advanced methodologies and agile project management techniques to manage software projects effectively		
Student Name:	T. Subahan	Reg. No.:	192511121
Date:	7/10/26	Duration:	60 minutes

Code of Conduct

I T. Subahan (192511121) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

- 1. Understand the Problem Statement**
 - Explain the problem in your own words.
 - Identify the objectives of the proposed system.
 - State the scope of the problem.
- 2. Identify Key Stakeholders and Challenges**
 - Identify all the stakeholders involved in the system.
 - Explain the roles and responsibilities of each stakeholder.
 - Discuss the major challenges and issues faced by the stakeholders.
- 3. Analyze the Current Approach**
 - Describe the existing system or current process.
 - Identify its strengths and limitations.
 - Analyze the problems or gaps that need to be addressed.
- 4. Propose a Suitable Solution Using Software Engineering Principles**
 - Design a software-based solution for the given problem.
 - Describe the major functional modules of the proposed system.
 - Identify the functional and non-functional requirements.
 - Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.
- 5. Justify the Chosen Methodology/Framework**

- Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
- Justify why the selected methodology/framework is suitable for the proposed system.
- Explain how it supports successful project development and maintenance.

6. Discuss Expected Outcomes and Limitations

- Describe the expected benefits of the proposed solution.
- Explain how the solution addresses the identified challenges.
- Discuss potential risks, implementation challenges, and limitations.
- Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.
Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification	Selects the most suitable SDLC/model/fram	Suitable methodology selected with	Methodology selected but justification is	Outcomes, limitations, or future enhancements

(10%)	ework and provides strong technical justification.	adequate justification.	weak.	are missing; report lacks organization and clarity.

Criteria	Mark
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

VEHICLE SERVICE AND MAINTENANCE MANAGEMENT SYSTEM

1. Understand the Problem Statement

1.1 Problem Statement

Vehicle owners need to regularly maintain their vehicles to ensure safety, reliability, and better performance. However, many vehicle owners currently depend on manual methods such as phone calls, paper records, service cards, or personal reminders to manage their vehicle servicing. These methods can result in missed service dates, loss of service history, difficulty in tracking ongoing repairs, and poor communication between customers and service centers.

The **Vehicle Service and Maintenance Management System** is proposed as a software-based solution to simplify and automate vehicle servicing activities. The system allows vehicle owners to register their vehicles, book service appointments, maintain service history, receive maintenance reminders, and track the current status of their vehicle service.

The system also helps service center staff manage appointments, update service progress, maintain vehicle records, and communicate service status to vehicle owners.

1.2 Objectives of the Proposed System

The main objectives of the Vehicle Service and Maintenance Management System are:

1. To provide an easy online platform for vehicle owners to book service appointments.
2. To maintain complete and organized vehicle service history.
3. To send reminders for upcoming services and maintenance activities.
4. To allow customers to track the current service status of their vehicles.
5. To help service center staff manage appointments efficiently.

6. To reduce manual paperwork and record-keeping.
7. To improve communication between vehicle owners and service center staff.
8. To provide accurate and easily accessible service information.
9. To improve customer satisfaction through faster and more transparent service management.
10. To provide a secure and reliable system for storing vehicle and service information.

1.3 Scope of the Problem

The proposed system covers the major activities involved in vehicle service and maintenance management.

The scope includes:

- User registration and login.
- Vehicle registration and management.
- Service appointment booking.
- Appointment rescheduling and cancellation.
- Service history management.
- Maintenance and service reminders.
- Service status tracking.
- Service advisor/technician updates.
- Customer notifications.
- Service center administration.
- Reports and service records.
- Secure storage of customer and vehicle information.

The system can initially be designed for a single service center and later expanded to support multiple service centers and different types of vehicles such as cars, bikes, and commercial vehicles.

2. Identify Key Stakeholders and Challenges

2.1 Key Stakeholders

The major stakeholders involved in the Vehicle Service and Maintenance Management System are:

- 1. Vehicle Owners/Customers**
- 2. Service Center Manager**
- 3. Service Advisors**
- 4. Service Technicians/Mechanics**
- 5. System Administrator**
- 6. Management/Service Center Owner**

2.2 Roles and Responsibilities

2.2.1 Vehicle Owner/Customer

The vehicle owner is the primary user of the system.

Responsibilities:

- Register and log in to the system.
- Add vehicle details.
- Book service appointments.
- Select required service types.
- View previous service records.
- Track service status.
- Receive maintenance reminders.
- Reschedule or cancel appointments.
- View completed service details.

2.2.2 Service Center Manager

The service center manager supervises the overall service operations.

Responsibilities:

- Monitor appointments.
- Assign vehicles to service staff.
- Monitor service progress.
- Manage service schedules.
- Review service records.
- Generate reports.
- Handle customer complaints and service-related issues.

2.2.3 Service Advisor

The service advisor acts as a communication link between the customer and service center.

Responsibilities:

- Confirm service appointments.
- Check vehicle service requirements.
- Update service information.
- Communicate estimated service details.
- Inform customers about additional maintenance requirements.
- Update service status.

2.2.4 Service Technician/Mechanic

The technician performs the actual inspection and repair of the vehicle.

Responsibilities:

- Inspect the vehicle.
- Identify maintenance or repair requirements.
- Perform assigned service activities.
- Update repair/service progress.
- Record replaced parts and completed work.

- Mark service activities as completed.

2.2.5 System Administrator

The administrator manages the software system.

Responsibilities:

- Manage user accounts.
- Manage service center information.
- Manage system settings.
- Monitor system security.
- Manage user access permissions.
- Maintain system data and backups.

2.2.6 Management/Service Center Owner

Management uses the system to monitor business and service performance.

Responsibilities:

- Monitor service center performance.
- Analyze service records.
- View reports.
- Monitor customer satisfaction.
- Make decisions based on service data.

2.3 Major Challenges Faced by Stakeholders

The major challenges include:

- Manual appointment scheduling can cause conflicts.
- Customers may forget their service dates.
- Paper-based service records can be lost or damaged.
- Customers may not know the current status of their vehicle.
- Service staff may have difficulty managing multiple appointments.
- Communication between customers and technicians may be inefficient.

- Maintaining large numbers of service records manually is difficult.
- Lack of centralized information can cause delays.
- Unauthorized access may expose customer and vehicle information.
- Customers may have to repeatedly contact the service center to obtain updates.

3. Analyze the Current Approach

3.1 Existing System/Current Process

In many traditional service centers, vehicle owners book appointments through phone calls, direct visits, or messaging applications. Service details are often recorded manually using paper forms, spreadsheets, or separate computer files.

A typical process is:

Customer → Phone/Direct Visit → Appointment Booking → Vehicle Inspection → Service/Repair → Manual Record Update → Customer Collection

The customer usually has to contact the service center to know whether the vehicle service has been completed.

3.2 Strengths of the Existing System

The existing approach has some advantages:

- Simple to understand.
- Does not require advanced technology.
- Suitable for small service centers.
- Direct communication between customer and service staff is possible.
- Initial implementation cost can be low.

3.3 Limitations of the Existing System

However, the traditional approach has several limitations:

1. **Manual booking:** Appointment booking depends heavily on phone calls or physical visits.
2. **Paper records:** Service records may be lost or damaged.
3. **No automatic reminders:** Customers may forget their next service date.
4. **Poor service tracking:** Customers cannot easily track service progress.
5. **Appointment conflicts:** Multiple customers may be assigned the same time slot.
6. **Data duplication:** The same customer or vehicle information may be entered repeatedly.
7. **Limited accessibility:** Records may not be available outside the service center.
8. **Communication delays:** Customers need to contact staff for updates.
9. **Difficult reporting:** Preparing service reports manually takes time.
10. **Security issues:** Physical records can be accessed or modified without proper authorization.

3.4 Problems/Gaps to Be Addressed

The proposed system addresses the following gaps:

Existing Problem	Proposed Solution
Manual appointment booking	Online appointment booking
Lost paper records	Digital service history
Missed service dates	Automatic reminders
No service tracking	Real-time service status
Appointment conflicts	Centralized scheduling
Repeated data entry	Stored customer and vehicle profiles
Communication delays	Automated notifications

Existing Problem

Difficult reporting

Unauthorized access

Proposed Solution

Digital reports and dashboards

Authentication and role-based access

4. Proposed Solution Using Software Engineering Principles

4.1 Proposed Software Solution

The proposed solution is a **web-based or mobile-based Vehicle Service and Maintenance Management System** that connects vehicle owners with service center staff.

The system stores customer, vehicle, appointment, service, maintenance, and service history information in a centralized database.

The basic working process is:

Customer Registration → Vehicle Registration → Service Booking → Appointment Confirmation → Vehicle Inspection → Service/Repair → Status Updates → Service Completion → Service History → Maintenance Reminder

The system provides different interfaces based on user roles. Customers can manage their vehicles and appointments, while service center staff can manage service operations.

4.2 Major Functional Modules

Module 1: User Registration and Login

This module manages customer and staff authentication.

Functions:

- New user registration.
- Login and logout.
- Password management.

- Role-based access.
- User profile management.

Module 2: Vehicle Management

This module stores vehicle information.

Vehicle details may include:

- Vehicle registration number.
- Vehicle type.
- Brand.
- Model.
- Manufacturing year.
- Current mileage.
- Fuel type.
- Vehicle owner details.

Module 3: Service Appointment Management

This module allows customers to book service appointments.

Functions:

- Select vehicle.
- Select service type.
- Select preferred date and time.
- Check available slots.
- Book appointment.
- Reschedule appointment.
- Cancel appointment.
- Receive appointment confirmation.

Module 4: Service History Management

This module maintains the complete service history of each vehicle.

It can store:

- Service date.
- Service type.
- Problems identified.
- Repairs performed.
- Parts replaced.
- Service cost.
- Vehicle mileage.
- Technician details.

Module 5: Maintenance Reminder

This module automatically reminds users about upcoming maintenance.

Examples include:

- Next regular service.
- Oil change.
- Brake inspection.
- Tyre replacement/check.
- Battery check.
- Insurance/vehicle-related reminders.

Module 6: Service Status Tracking

This module allows customers to monitor their vehicle's service progress.

Possible statuses include:

Appointment Confirmed → Vehicle Received → Inspection → Service in Progress → Quality Check → Ready for Delivery → Completed

Module 7: Notification Module

This module sends notifications to customers.

Notifications may include:

- Appointment confirmation.
- Appointment reminder.
- Service started.
- Additional repair requirement.
- Service completed.
- Vehicle ready for collection.
- Upcoming maintenance reminder.

Module 8: Admin and Service Center Management

The administrator and manager can:

- Manage users.
- Manage technicians.
- Manage appointments.
- Manage service types.
- Assign technicians.
- Monitor service status.
- Generate reports.
- Manage system settings.

4.3 Functional Requirements

Functional requirements describe what the system should do.

The proposed system should:

1. Allow users to create accounts.
2. Allow users to securely log in.
3. Allow users to add and manage vehicle details.
4. Allow customers to book service appointments.
5. Allow customers to reschedule or cancel appointments.

6. Allow staff to approve and manage appointments.
7. Allow technicians to update service status.
8. Maintain vehicle service history.
9. Generate maintenance reminders.
10. Send notifications to customers.
11. Allow customers to track service status.
12. Allow administrators to manage users and services.
13. Generate service-related reports.
14. Maintain secure records of all service activities.

4.4 Non-Functional Requirements

1. Usability

The system should have a simple and user-friendly interface so that customers can easily book appointments and check service status.

2. Reliability

The system should operate consistently and ensure that service and appointment information is not lost.

3. Security

Customer information, login credentials, vehicle information, and service records should be protected using authentication and authorization mechanisms.

4. Performance

The system should respond quickly when users book appointments, view service history, or track service status.

5. Scalability

The system should be capable of supporting an increasing number of customers, vehicles, appointments, and service centers.

6. Maintainability

The software should be modular so that individual components can be modified or updated without affecting the complete system.

7. Availability

The system should be available whenever customers need to book appointments or check service information.

8. Data Integrity

The system should ensure that customer, vehicle, appointment, and service records remain accurate and consistent.

4.5 Application of Software Engineering Principles

Modularity

The system is divided into independent modules such as user management, vehicle management, appointment management, service history, reminders, and notifications. This makes development and maintenance easier.

Scalability

The system can initially support a single service center and later be expanded to support multiple service centers and thousands of users.

Maintainability

A modular design allows developers to modify or upgrade individual modules without changing the entire system.

Reliability

Database backup, validation, error handling, and proper transaction management can be used to prevent data loss and system failures.

Usability

A simple interface with clear navigation, buttons, forms, notifications, and service

status indicators makes the system easy to use.

Security

The system can implement:

- Username/password authentication.
- Role-based access control.
- Encrypted passwords.
- Secure database access.
- Input validation.
- Regular data backups.

5. Justify the Chosen Methodology/Framework

5.1 Selected SDLC Model: Agile Methodology

The **Agile Software Development Methodology** is suitable for developing the Vehicle Service and Maintenance Management System.

Agile follows an iterative and incremental approach in which the software is developed in small parts called iterations or sprints.

5.2 Why Agile is Suitable

Agile is suitable for this project because the requirements of customers and service centers may change during development.

For example, after seeing the initial system, service center staff may request additional features such as:

- Online payment.
- Invoice generation.
- Spare-parts tracking.
- Customer feedback.
- WhatsApp/SMS notifications.

- Pickup and delivery services.

Agile allows these requirements to be added in later development cycles.

5.3 Agile Development Process

The project can be developed using the following stages:

Requirement Analysis → Sprint Planning → Design → Development → Testing → Customer Feedback → Improvement → Next Sprint

Sprint 1

Develop:

- User registration.
- Login.
- User profile.

Sprint 2

Develop:

- Vehicle registration.
- Vehicle management.

Sprint 3

Develop:

- Appointment booking.
- Appointment management.

Sprint 4

Develop:

- Service history.
- Service status tracking.

Sprint 5

Develop:

- Maintenance reminders.

- Notifications.

Sprint 6

Develop:

- Admin dashboard.
- Reports.
- Final testing.

5.4 Benefits of Agile for This Project

- Supports changing requirements.
- Provides continuous customer feedback.
- Allows early delivery of working modules.
- Makes testing continuous.
- Helps identify problems early.
- Reduces development risks.
- Improves software quality.
- Supports continuous maintenance and enhancement.

6. Expected Outcomes and Limitations

6.1 Expected Outcomes

The proposed system is expected to provide the following benefits:

1. Easy and convenient service appointment booking.
2. Reduced manual paperwork.
3. Organized digital service history.
4. Automatic maintenance reminders.
5. Improved communication with customers.
6. Better service status visibility.
7. Reduced appointment conflicts.

8. Faster service management.
9. Improved customer satisfaction.
10. Better management decision-making through reports.
11. Improved data security.
12. Reduced chances of missing scheduled maintenance.

6.2 How the Solution Addresses the Challenges

Challenge	System Solution
Missed service appointments	Automatic reminders
Manual records	Digital database
Long waiting time	Online appointment scheduling
Lack of service updates	Service status tracking
Lost service history	Centralized digital records
Appointment conflicts	Automated scheduling
Communication problems	Notifications
Difficult management	Admin dashboard and reports
Data security problems	Authentication and access control

6.3 Potential Risks

Some risks associated with the system include:

- Server or database failure.
- Internet connectivity problems.
- Unauthorized access.
- Incorrect data entry.
- Notification delivery failure.
- Software bugs.
- User resistance to adopting the new system.

- Data privacy concerns.
- High initial development and implementation cost.

6.4 Implementation Challenges

The major implementation challenges include:

1. Designing a simple interface for different types of users.
2. Migrating existing paper records into the digital system.
3. Training service center staff.
4. Maintaining accurate service information.
5. Integrating notification services.
6. Protecting customer information.
7. Ensuring system availability.
8. Integrating the system with existing service center software, if available.

6.5 Limitations

The proposed system may have some limitations:

- It depends on internet connectivity for online access.
- Initial implementation requires time and resources.
- Staff may require training.
- Automatic reminders depend on correct vehicle and service data.
- Integration with external payment or notification services may increase complexity.
- The first version may not support all vehicle manufacturers or specialized service requirements.

6.6 Future Enhancements

The system can be enhanced in the future with the following features:

1. Online Payment

Customers can pay service bills through integrated digital payment systems.

2. Online Invoice Generation

The system can automatically generate and provide digital invoices after service completion.

3. Mobile Application

A dedicated Android/iOS application can provide easier access to service features.

4. AI-Based Predictive Maintenance

Artificial Intelligence can analyze vehicle service history and predict possible maintenance requirements before a failure occurs.

5. IoT Integration

IoT sensors can collect real-time vehicle information and send it to the system for monitoring.

6. Spare Parts Management

The system can maintain information about available spare parts, inventory levels, and replacement history.

7. Customer Feedback and Rating

Customers can provide ratings and feedback about service quality.

8. Pickup and Delivery Service

Customers can request vehicle pickup and delivery from their location.

9. Multi-Service-Center Support

The system can be expanded to support multiple branches and allow customers to select their preferred service center.

10. AI Chatbot

An AI chatbot can answer common questions related to service schedules, maintenance, appointments, and vehicle care.

Conclusion

The **Vehicle Service and Maintenance Management System** provides a software-based solution for managing vehicle servicing activities in an efficient and organized manner. The system replaces many manual processes with digital appointment booking, service history management, maintenance reminders, notifications, and service status tracking.

The use of **Agile methodology** makes the system suitable for iterative development and allows feedback and changing requirements to be incorporated throughout the project. By applying software engineering principles such as modularity, scalability, maintainability, usability, reliability, and security, the proposed system can provide a reliable and user-friendly solution for modern vehicle service management.